The Solar System: A Short Guide

The Solar System formed about 4.6 billion years ago from a collapsing cloud of gas and dust. The Sun contains more than 99 percent of the total mass of the system. Eight planets orbit the Sun: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

The inner planets, Mercury, Venus, Earth and Mars, are rocky. The outer planets are much larger. Jupiter and Saturn are gas giants made mostly of hydrogen and helium, while Uranus and Neptune are ice giants rich in water, ammonia and methane.

Mercury and Venus

Mercury is the smallest planet and the closest to the Sun. It completes an orbit in just 88 Earth days. It has almost no atmosphere, so temperatures swing from about 430 degrees Celsius by day to minus 180 at night.

Venus is the hottest planet, with a surface temperature of roughly 465 degrees Celsius, because its thick carbon dioxide atmosphere traps heat in a runaway greenhouse effect. A day on Venus is longer than its year, and it rotates in the opposite direction to most planets.

Earth and Mars

Earth is the only known world with liquid water on its surface and with life. Its single moon stabilises the tilt of its axis. Mars is about half the diameter of Earth and has two small moons, Phobos and Deimos.

Mars shows the red colour of iron oxide dust. It hosts Olympus Mons, the tallest volcano in the Solar System at about 22 kilometres high, and Valles Marineris, a canyon system more than 4,000 kilometres long. NASA's Perseverance rover landed in Jezero Crater in February 2021 to search for signs of ancient microbial life.

Jupiter and Saturn

Jupiter is the largest planet. Its mass is more than twice that of all the other planets combined. The Great Red Spot is a giant storm that has been observed for centuries. Jupiter has at least 90 known moons, including Ganymede, the largest moon in the Solar System.

Saturn is famous for its rings, which are made of ice particles and rock. Saturn is the least dense planet, less dense than water. Its moon Titan has a thick atmosphere and lakes of liquid methane. The Cassini spacecraft orbited Saturn from 2004 until 2017.

Uranus and Neptune

Uranus rotates on its side, with an axial tilt of about 98 degrees. It was discovered by William Herschel in 1781. Neptune was found in 1846 after mathematicians predicted its position from irregularities in the orbit of Uranus.

Neptune has the strongest winds of any planet, reaching more than 2,000 kilometres per hour. Voyager 2 is the only spacecraft to have visited both Uranus and Neptune, flying past them in 1986 and 1989.